UAV Disaster Surveillance Strategy Engine

Genetic Algorithm Optimization — Artificial Intelligence Assignment

Team members: [Enter all three names] **Date:** [Enter submission date]

1. Introduction and problem formulation

This project implements an educational strategy engine for three simulated UAVs surveying a 50 × 50 grid containing priority targets, no-fly zones, and graded hazard regions. The task balances target coverage and detection against energy expenditure, flight time, and hazard exposure. A Genetic Algorithm (GA) evolves a population of candidate routes using selection, crossover, mutation, route repair, and elitism. A GA is a stochastic heuristic and does not guarantee a globally optimal solution.

2. Chromosome design

Each chromosome contains one plan per UAV. Each plan includes eight waypoint coordinate pairs, eight normalized speed values in [0.2, 1.0], and eight binary sensor activation values. Segment crossover swaps a waypoint interval along with matching speed and sensor genes. Mutation perturbs spatial coordinates, changes speed, or flips a sensor bit. Repair is applied after genetic operations to keep coordinates on the grid and avoid blocked route segments.

3. Fitness function and constraints

F = 0.35C + 0.25D − 0.15E − 0.10T − 0.15H − P

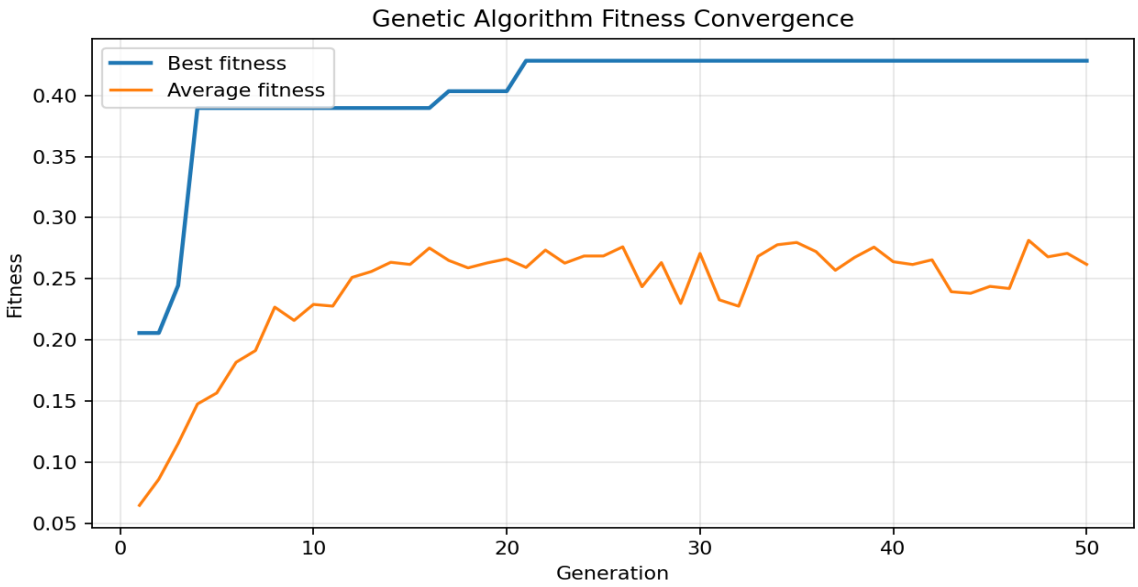
C is weighted coverage, D is sensor-enabled target detection, E is normalized energy, T is normalized flight time, H is normalized hazard exposure, and P penalizes route-range or no-fly violations. Coverage and detection receive the largest positive weights because mission success is prioritized. Energy is estimated from distance, speed, and sensor activation; time is approximated from distance and speed; hazard exposure accumulates severity along sampled route cells. These are simplified coursework assumptions, not real flight dynamics. The best candidate is also checked against range, battery, and no-fly constraints.

4. Experimental setup and main results

The main run used population 30, 50 generations, crossover probability 0.8, mutation rate 0.15, tournament size 3, and elitism of 2. The GA and random search each evaluated 2536 candidates. The GA best fitness was **0.4293**, versus **0.3559** for random search. The best candidate's weighted coverage/detection was 0.956, estimated energy 422.84, estimated time 358.57, and hazard exposure 15.75. The validator reported no range, battery, or no-fly violations for the best candidate.

Metric	Observed result
GA best fitness	0.4293
Random-search best fitness	0.3559
Weighted coverage / detection	0.956
Estimated energy	422.84
Estimated flight time	358.57
Hazard exposure	15.75
Evaluations per method	2536

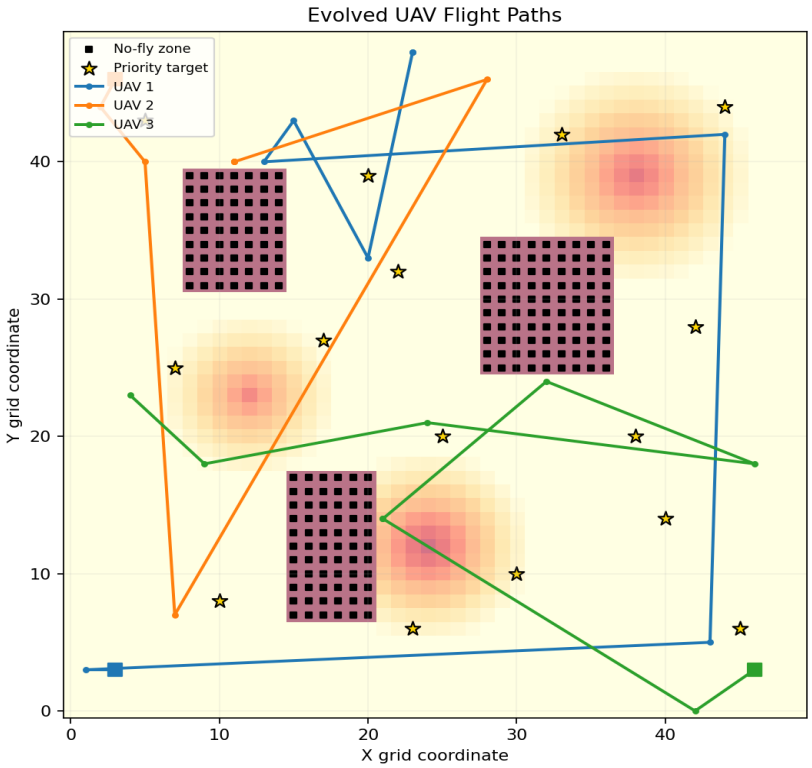
Fitness convergence


Genetic Algorithm Fitness Convergence
Best fitness
Average fitness
Fitness
Generation
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
0
10
20
30
40
50

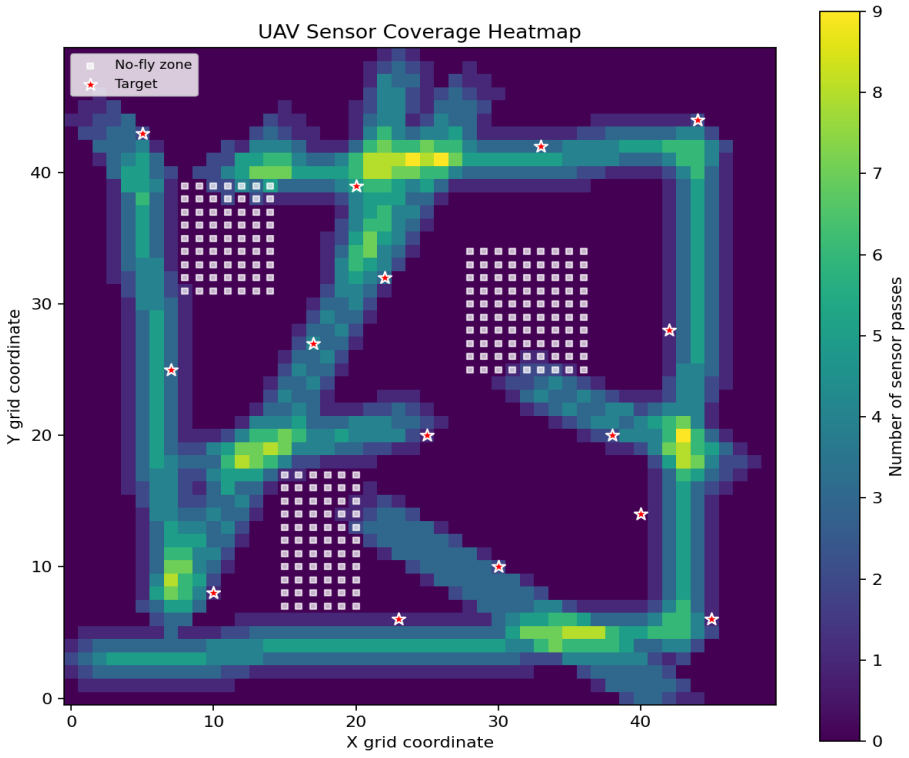
5. Evolved routes and coverage

The route plot overlays UAV trajectories with targets, no-fly cells, and hazard severity. The coverage heatmap counts sensor-active route passes. These plots show the spatial pattern of the candidate solution; they should be interpreted alongside the numeric fitness and feasibility checks.

Evolved UAV flight paths

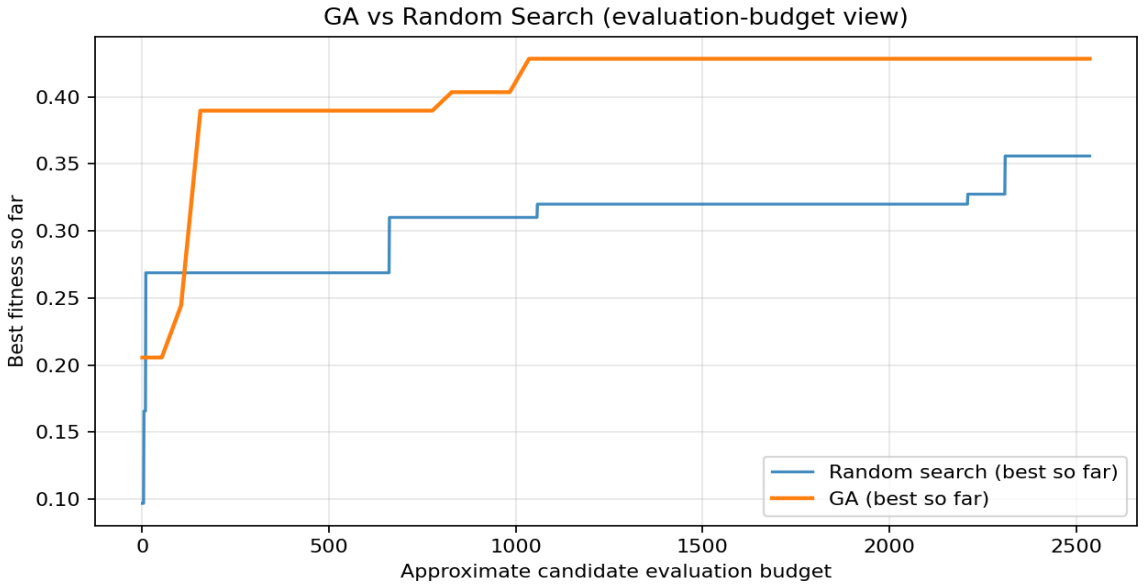


Sensor coverage heatmap



6. GA comparison and hyperparameter sensitivity

Under the same candidate-evaluation budget, the GA produced a higher best fitness than random search in this run. This is evidence for this particular seeded experiment only; repeated trials are needed to establish a robust advantage.



The short sensitivity experiments varied population size, mutation rate, and tournament size. Each configuration used five generations, so these values are preliminary and may vary with random seed.

Configuration	Best fitness
pop=10	0.2902
pop=15	0.3560
pop=20	0.2576
mut=0.05	0.2956
mut=0.15	0.3766
mut=0.30	0.2818
tour=2	0.2628
tour=3	0.3014
tour=5	0.3556

Within this short experiment, the best population-size configuration was pop=15 (fitness 0.3560), the best mutation configuration was mut=0.15 (fitness 0.3766), and the best tournament configuration was tour=5 (fitness 0.3556). Larger populations cost more computation; low mutation can limit exploration, high mutation can disrupt useful routes, and stronger tournament selection can reduce diversity.

7. Conclusion and limitations

The project demonstrates a modular GA that encodes waypoints, speed, and sensor schedules, evaluates a weighted multi-objective score, and uses repair and validation for basic constraints. The tested GA outperformed random search under the same evaluation count. Limitations include simplified energy/time models, static hazards, heuristic repair, and fixed waypoint count. This is a coursework prototype, not a planner for real-world UAV deployment. For stronger evidence, repeat each sensitivity configuration across several seeds and report means, standard deviations, feasibility rates, and runtime.

References: add the course notes, textbooks, or external sources actually consulted. Replace the team-member and date placeholders before submission.